

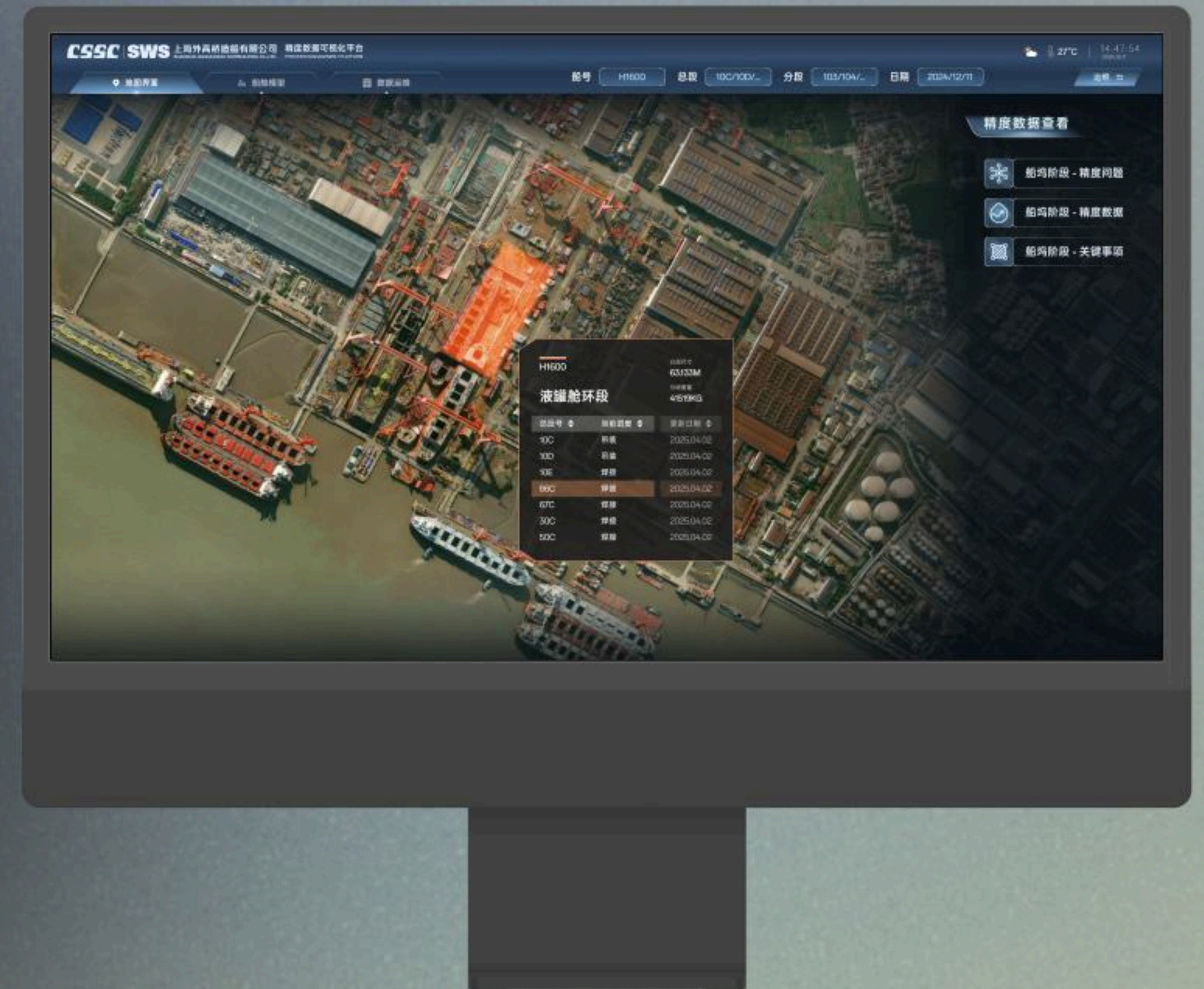
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Precision Management System for Shipbuilding

UI/UX Design

Industrial Digitalization

ToB



Client's Request

"We need a 3D-enabled precision management system that helps teams track shipbuilding progress, inspect precision issues, and maintain operational data in one unified environment."



My Role Internship @ Unity®

As a Product Intern and the **sole UI/UX designer on the project** at Unity, I joined the project at a very early stage, when the team only had a technical agreement and an initial PoC plan. I was responsible for shaping and delivering the product content defined in the **PoC** scope.

Since shipbuilding was a completely new domain for me, I spent significant time learning the industry context, aligning repeatedly with the client, uncovering real operational pain points, and **conducting on-site field research at Shanghai Waigaoqiao Shipbuilding**. This process helped me translate technical requirements into a clearer, more grounded product direction.

Client	CSSC
Location	Shanghai
Year	2025
Duration	5 months

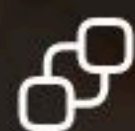
Problem Statement

In shipbuilding, progress records, precision issues, and engineering references are often distributed across separate documents, spreadsheets, and model files, making it difficult for teams to see the full picture in one place.



Scattered Records

Data was spread across spreadsheets and repeatedly shared by hand.



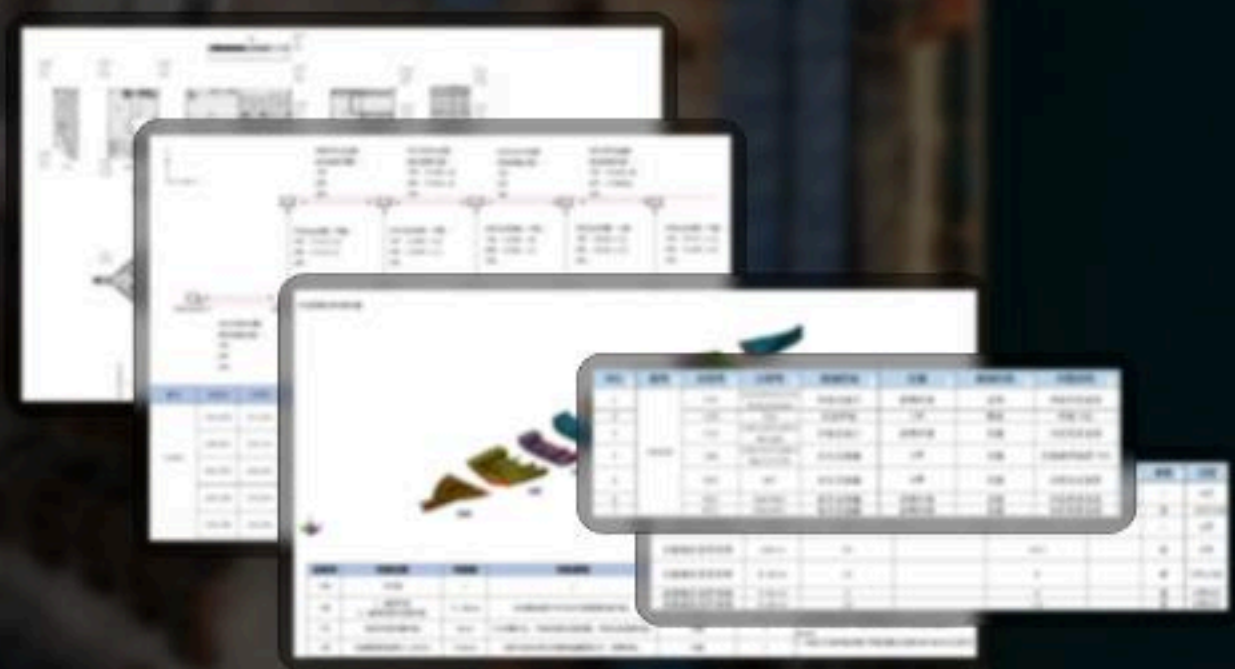
Manual Cross-Checking

Teams had to compare 2D drawings, 3D models, and tables manually.



Limited Annotation

Traditional documents made spatial issue marking difficult.



Our Strength Unity®

Unity Industry extends the Unity engine for industrial workflows.

It combines real-time 3D interaction with tools for importing CAD and other complex 3D data, building custom applications, and deploying them across platforms.



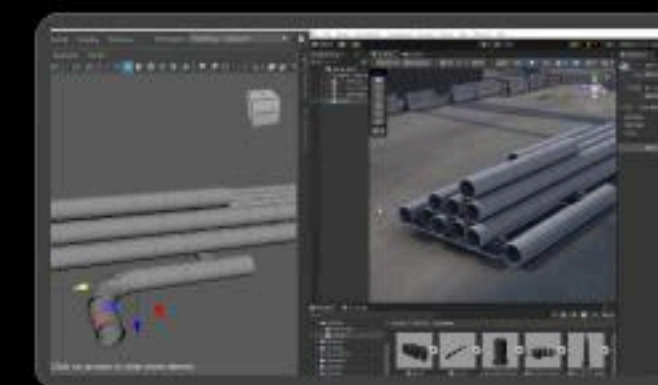
Connected Data

Brings CAD and 3D assets into one connected workflow.



Interactive Context

Turns technical records into explorable 3D applications.



Scalable Workflow

Supports custom industrial applications from data integration to deployment.



Three Information Layers, One Workflow

Users needed to move across these layers fluidly, switching between them to understand each construction or precision issue from multiple perspectives.

1 Dockyard Layer

Shows how construction is progressing across platforms and docks.

Search by ship / block

View progress

2 Ship Model Layer

Makes precision information easier to understand in spatial context.

Select blocks

Highlight segments

Compare lifting status

3 Data Layer

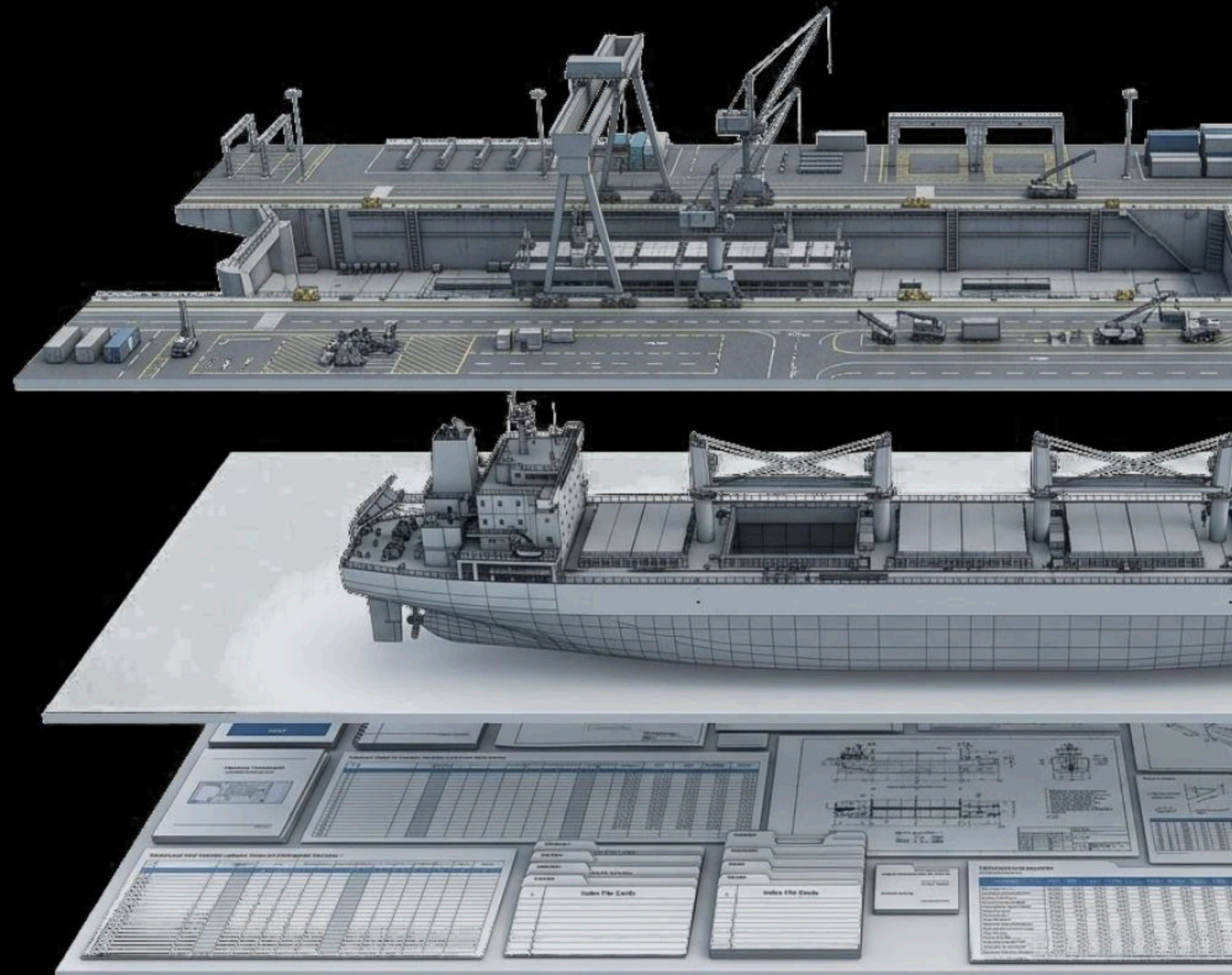
Supports both read-only inspection and editable maintenance workflows.

Edit records

Update status

Add 3D labels

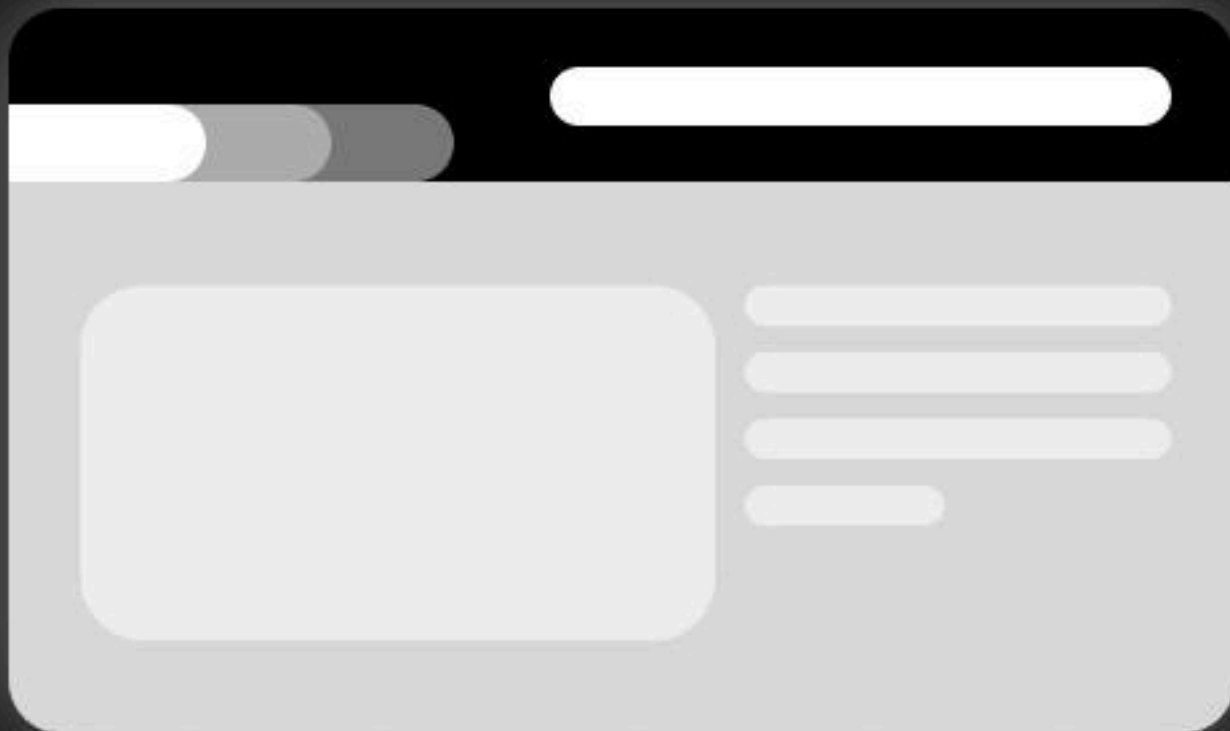
Draw cut lines



Orientation Across Layers

To help users navigate complex ship structures and layered information, the interface provided continuous orientation cues across the workflow. From the current block and active layer, to the model structure tree and contextual detail popups, each element helped users stay grounded while moving between overview and detail.

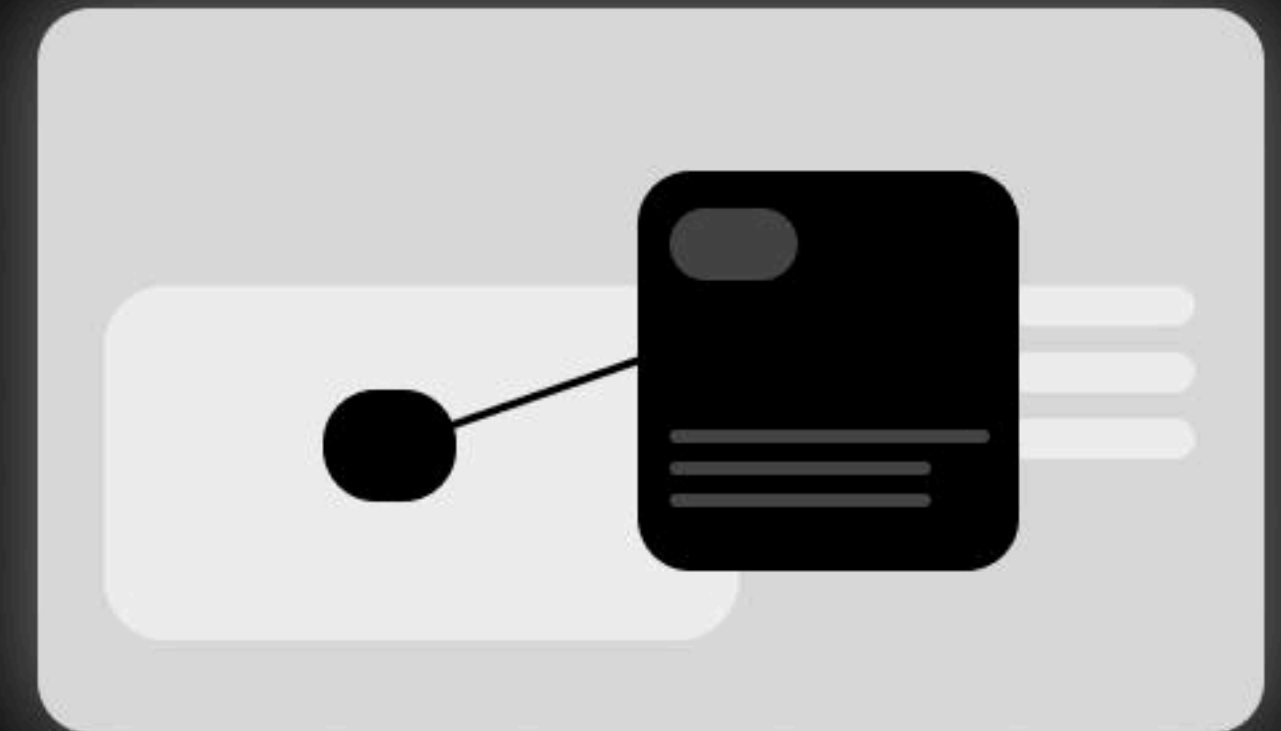
Global Context

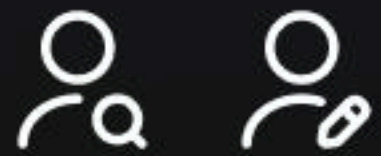


Structure Tree



Contextual Popups



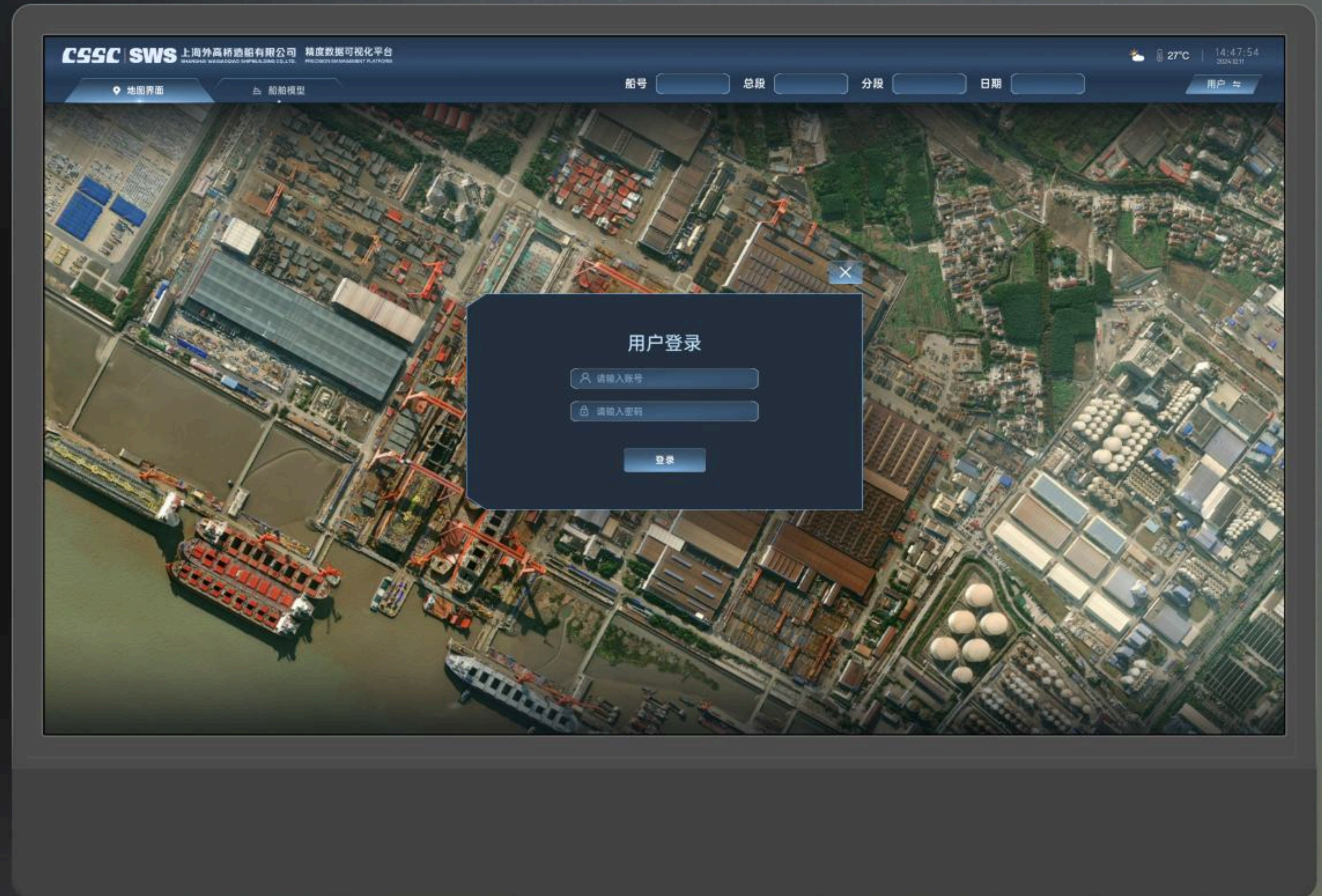


Role-Based Entry Point

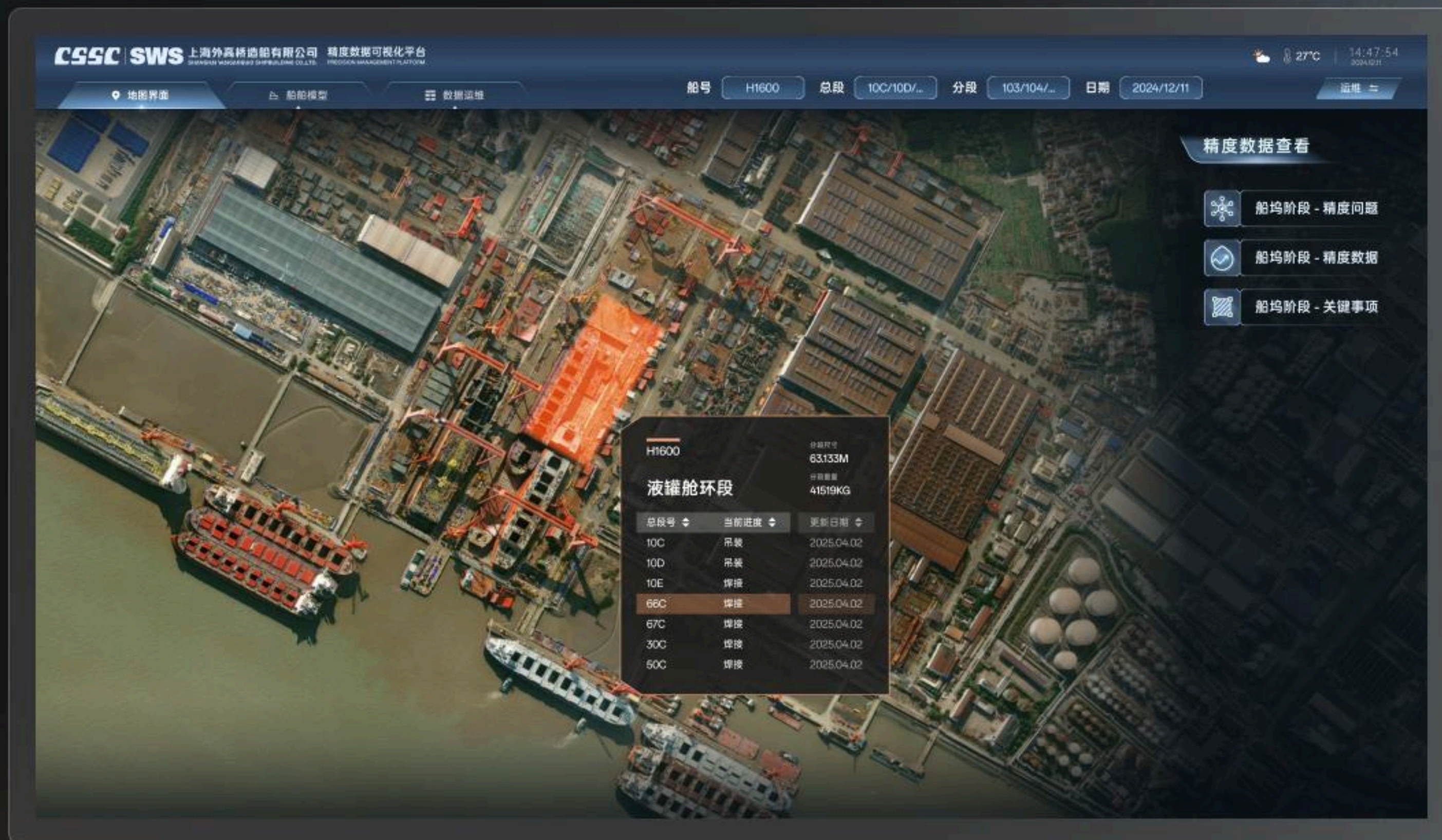
Management users mainly inspect progress and precision conditions, while operations users need additional access to editing and maintenance tasks.

Management User

Operations User

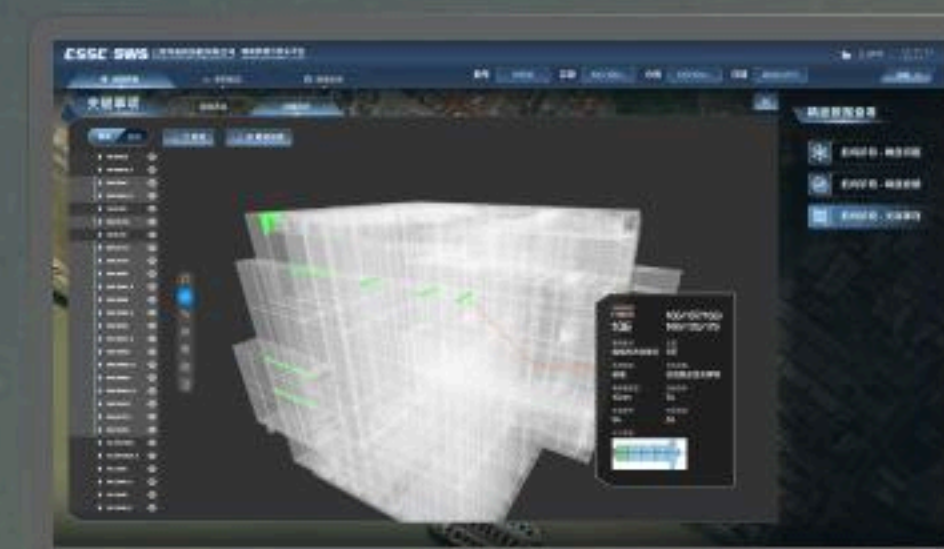
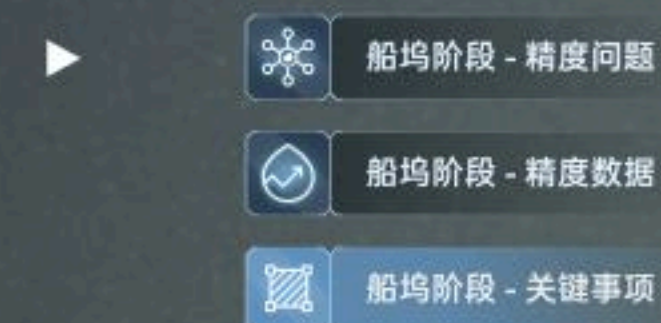


Dockyard View



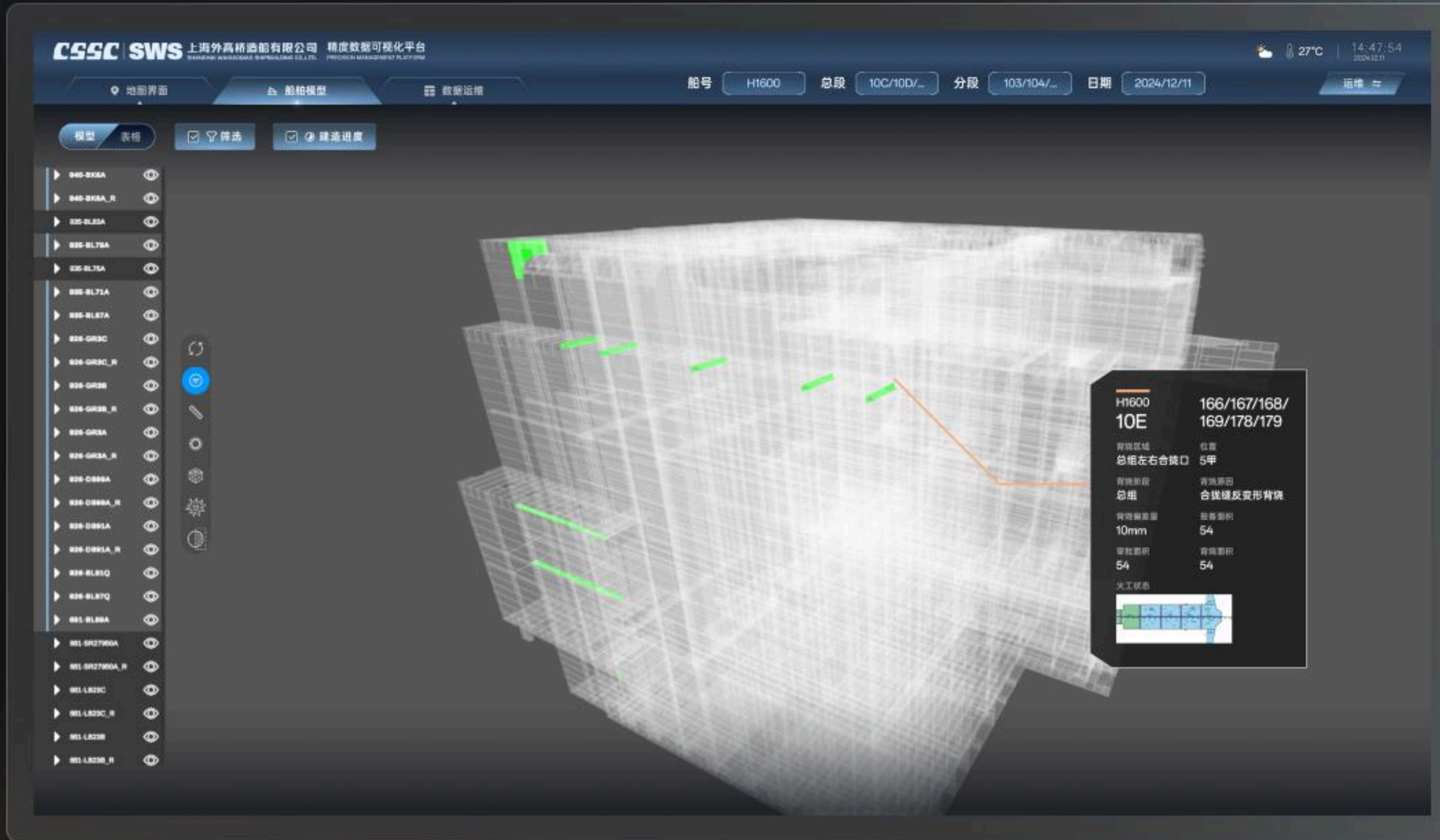
Filter by **ship, block, and date** to locate a section in the dockyard. The map then auto-navigates and highlights the relevant area.

After selecting a section, users can preview different types of precision data from the right-side panel in a **pop-up window**.





Ship Model View



Users can navigate the model through the structure tree and apply additional filters to narrow the view. Precision markings appear directly in 3D, while a pop-up reveals key details for the selected item.



Backburning Region: 总组左右合拢口

Backburning Position: 一甲/二甲/三甲/机舱甲板

Backburning Stage: 总组/搭载/码头/其他

Backburning Reason: 合拢缝反变形背烧

Backburning Deviation: -20mm to 5mm

Reporting Area: -20m² to 50m²

Approval Area: -20m² to 50m²

Backburning Area: -20m² to 50m²

取消 确认

☐ 总段切割情况 +

☐ 复位情况 +

☐ 定位数据 +

☐ 立柱数据 +

☒ 背烧统计 -

Backburning Region: 总组左右合拢口

Backburning Position: 一甲/二甲/三甲/机舱甲板

Backburning Stage: 总组/搭载/码头/其他

Backburning Reason: 甲板平整度不良

Backburning Deviation: -20mm to 5mm

Reporting Area: -20m² to 50m²

取消 确认

☐ 总段切割情况 +

☐ 复位情况 +

☐ 定位数据 +

☐ 立柱数据 +

☒ 背烧统计 +

取消 确认



Data View

CSSC SWS 上海外高桥造船有限公司 精度数据可视化平台
SHANGHAI WAIGAOQIAO SHIPBUILDING CO., LTD. PRECISION MANAGEMENT PLATFORM

地图界面

船舶模型

数据运维

船号 H1600 总段 10C/10D/... 分段 103/104/... 日期 2024/12/11 运维

添加事项

建造进度

总段切割情况

搭载就位情况

搭载定位数据

立柱数据

背烧统计

+

序号	船号	总段号	分段号	背烧区域	背烧位置	背烧阶段	背烧原因	偏差量	报备面积	审批面积	背烧面积	火工状态	查看模型	更新时间
1	H1600	10C	103/104/143/153/124/154	外板合拢口	舷侧外板	总组	合拢缝反变形背烧	5-8mm	85	85	85	图片	模型	2025.03.24
2	H1600	10D	105	内部甲板	1甲	搭载	平整度不良背烧	-12mm	10	10	10	图片	模型	2025.03.24
3	H1600	10D	105/125/135/145/155	外板合拢口	舷侧外板	总组	合拢缝反变形背烧	3-8mm	50	50	50	图片	模型	2025.03.24
4	H1600	10E	166/167/168/169/17/179	左右总组缝	5甲	总组	合拢缝反变形背烧	10mm	54	54	54	图片	模型	2025.03.24
5	H1600	30C	347	左右总组缝	8甲	总组	合拢缝反变形背烧	6-8mm	12	12	12	图片	模型	2025.03.24
6	H1600	66C	644/645	前后总组缝	舷侧外板	总组	合拢缝反变形背烧	6-8mm	8	8	8	图片	模型	2025.03.24
7	H1600	67C	654/655	前后总组缝	舷侧外板	总组	合拢缝反变形背烧	6-8mm	10	10	10	图片	模型	2025.03.24

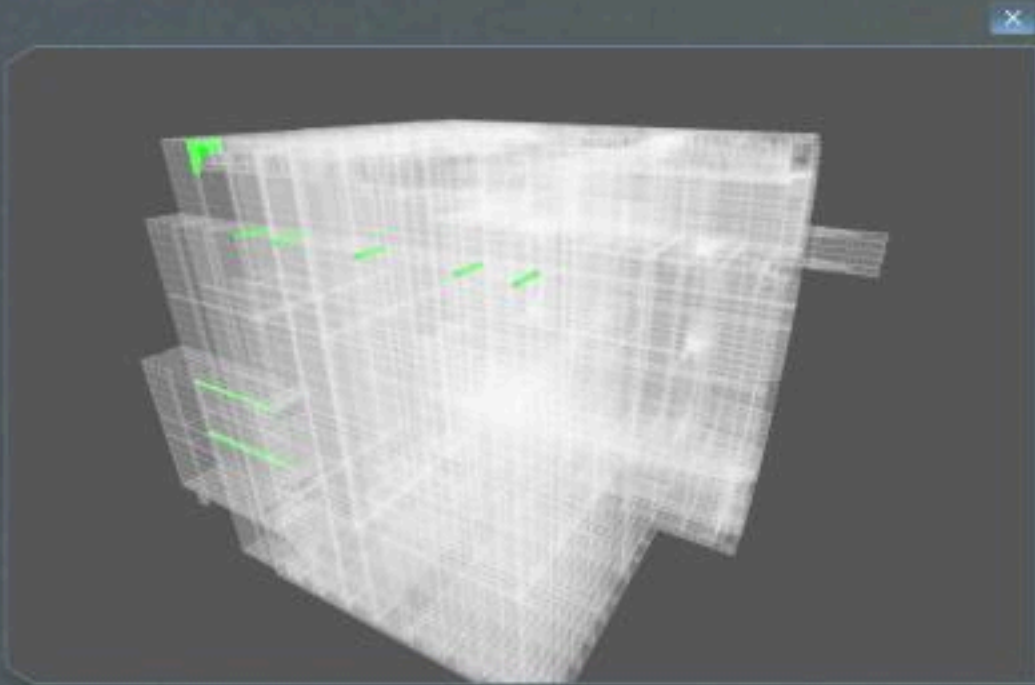
Progress Tracking

View each section's current construction stage.



3D Markup Preview

Inspect model annotations directly in 3D.



Record Management

Create, edit, and manage entries in table view.

请选择或者新增选项

外板合拢口

内部甲板

左右总组缝

前后总组缝

五甲

五甲

请选择或者新增选项

外板合拢口

内部甲板

左右总组缝

前后总组缝

新增 五甲

五甲

五甲

请选择或者新增选项

外板合拢口

内部甲板

左右总组缝

前后总组缝

新增 五甲



Built with **WebGL** for lightweight 3D
access directly in the browser.

Component Library



Next Design Direction

The next step is to **adapt to other industrial scenarios** that also rely on spatial inspection, model-linked records, and precision control.

